LU Lebanese University

Team project defense · Topic: Grounding

DATA SCIENCE FOR CONVERSATIONAL AI · PAPER: REFUSALBENCH, EACL 2026

Reading, Not Asking

Can a small language model's inner signal tell it when to refuse? A reproduction and two extensions.

Abbass Zahreddine · MSc (M2), Faculty of Sciences · 12 minutes

THE PAPER · THE PROBLEM

A grounded model must say no when the evidence is bad.

THE PAPER · PEER REVIEWED

RefusalBench: Generative Evaluation of Selective Refusal in Grounded Language Models

Muhamed, Ribeiro, Dreyer, Smith, Diab · Carnegie Mellon and Amazon · EACL 2026 · arXiv 2510.10390

- **Grounded** answering: the model may use only the passage we give it.
- **Selective refusal**: answer if the passage supports it, refuse if not, and name the reason.

ORIGINAL · CLEAR

Q: When did the East India Company take control of India?
Passage: The East India Company gained a foothold in India in 1611.
Answer: 1611

PERTURBED · AMBIGUOUS

Q: When did *the company* take control of India?
Passage adds: The Dutch East India Company established posts in India.
Correct reply: `REFUSE_AMBIGUOUS_QUERY`

THE PAPER · HOW IT MEASURES, WHAT IT FOUND

It breaks good evidence on purpose, then tests two skills.

01	02	03	04
Start	**Break it**	**Check**	**Test**
Natural Questions with a good passage	176 rewrite rules, 6 failure types, 3 strengths	Several judge models must agree	1,600 examples: 536 answerable, 1,064 to refuse

Six failure types: **ambiguity**, **contradiction**, **missing information**, **false premise**, **wrong level of detail**, **opinion asked**.

WHAT THE PAPER FOUND

- Best refusal accuracy on this set: **73.0%** (Claude 4 Sonnet).
- On multi-document passages: **47.4%** at best.
- **Detecting** that a refusal is needed is easier than saying **why**.

Our model, Qwen1.5-7B-Chat: answer accuracy 56.1%, refusal accuracy about 5%.

BASELINE · WHAT WE REPRODUCED AND WHAT WE CHANGED

We reproduced Qwen1.5-7B on free resources, and wrote down every change.

METRIC	PAPER	OURS
Answer accuracy	0.561	**0.690**
Refusal accuracy (exact reason)	about 0.05	**0.114**
Answerable questions refused	not given	**0.165**
Needed refusals missed	not given	**0.456**

Same finding as the paper: a 7B model almost never names the right reason.

- Same data and **same prompt** as the paper.
- Qwen1.5-7B-Chat, **4-bit**, on a free Colab T4 GPU.
- Judge: **Gemini** (the paper used Claude). Clean refusal codes are scored by code, the rest by the judge.
- Sampling like the paper: temperature 1.0.

HONEST STATUS

1,560 of 1,600 replies generated (40 ran out of GPU memory). **770 graded so far.** Cost: $0.

EXTENSION 1 · THE LIMITATION AND AN IDEA THAT FAILED

Asking the model to diagnose first made it worse.

- **The limitation:** the baseline misses 46% of needed refusals, mostly ambiguity (0.74) and contradiction (0.71).
- **The idea:** make the model write `EVIDENCE_STATE` first, then answer.
- **The control:** only ask for a final line, to rule out formatting.

WHAT WE CLAIM

We report the result. We did not test why it happens.

	BASELINE	CONTROL	DIAGNOSE FIRST
Refusals caught (%)	43.3	35.8	**1.9**
Right reason (%)	6.8	3.3	**0.0**

95% of the time the model wrote CLEAR (762 of 800), even when it had to refuse.

EXTENSION 2 · THE IDEA AND HOW WE BUILT IT

Do not ask the model. Read it.

S1 · REFUSAL PROBABILITY

How likely is a refusal code?

The chance that the reply **starts** with a refusal code. No text is generated.

S2 · YES OR NO

Can this be answered?

Ask a yes or no question. Compare the chance of "No" with the chance of "Yes".

S3 · LINEAR PROBE

What is inside?

A small classifier trained on the model's **hidden state**, its inner numbers.

- **One rule for all:** refuse if the score is above a threshold.
- **Honest test:** thresholds and probe are tuned on other questions (5 folds over the 100 source questions).
- About 0.3 seconds per example on a T4. No judge. $0.

EXTENSION 2 · RESULTS AGAINST THE BASELINE

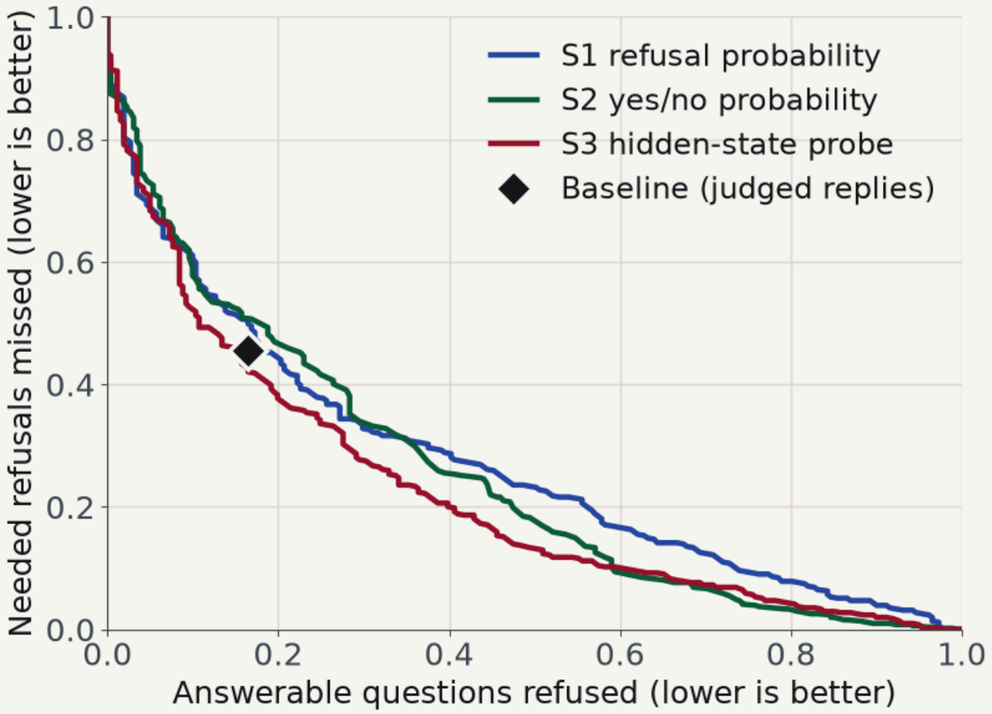
The signal is real, but a simple threshold does not beat the baseline.

READOUT	SIGNAL	DECISION	VS BASELINE
Baseline	none	**0.690**	reference
S1	0.726	0.658	slightly worse
S2	0.751	0.669	unclear
S3 probe	**0.797**	**0.718**	unclear

Signal = AUROC (0.5 coin flip, 1.0 perfect). **Decision** = balanced accuracy (0.5 chance). Probe gain: +0.028, interval includes 0.

LIMITS

770 of 1,560 graded · judge unchecked · one seed



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